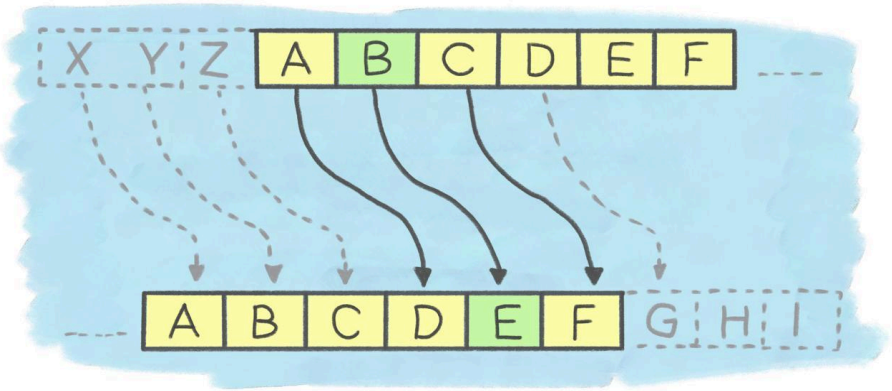
Disc Enigma

Introduction

The need for secrecy when sending messages over the Internet is important for everyone. With this activity, students are introduced to the concept of basic cryptography/encryption, and to simple techniques for writing encrypted messages. Here, we will understand the workings of the Enigma machine by building a simpler Disc Enigma and using it to encrypt and decrypt a message.



Activity	Time	Description
Launch	5 min	Start by asking students if they have ever shared a secret message with a friend. Ask questions like: How did you make sure others could not understand it? or Did you ever use code or secret language? Teacher can even write a sentence on the board which is encrypted, and the student has to guess the message. Tell students that people also needed to send secret messages long ago. Introduce the idea that, even today, computers use special codes to hide information and keep messages safe on the Internet. Explain that in this activity, they will learn about an Enigma machine, which was used in World War II and was nearly impossible to crack the code, and they will even make their own simpler version of the Enigma machine, the Disc Enigma, to send secret messages to their friends. Template: https://tinyurl.com/discenigmatemplate
Making & exploration of the Disc Enigma by Students	20 min	Show the Video Demonstration of Actual Enigma: https://tinyurl.com/discenigmavideo Discuss the complexity of the Enigma Machine.

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		Making the Disc Enigma Model and having students try out the activity together, following the worksheet questions. Student Worksheet: https://tinyurl.com/discenigmaworksheet
Discussions & Conclusion	10 min	Discussing the worksheets.

CT Components

Algorithmic Thinking
We follow a step-by-step process for encrypting and decrypting messages.

Decomposition
We break the task of sending a secret message into smaller steps: choosing a message, deciding the key pattern, encoding each letter, and decoding it later.

Pattern Recognition and Generalisation
Recognise that the mapping of letters follows a systematic pattern. By exploring the trivial case for 2 and 3 letters, this idea extends to the full alphabet.

Evaluation
Evaluate how secure the Disc Enigma really is by exploring that there are 26! possible key patterns to encrypt the message.

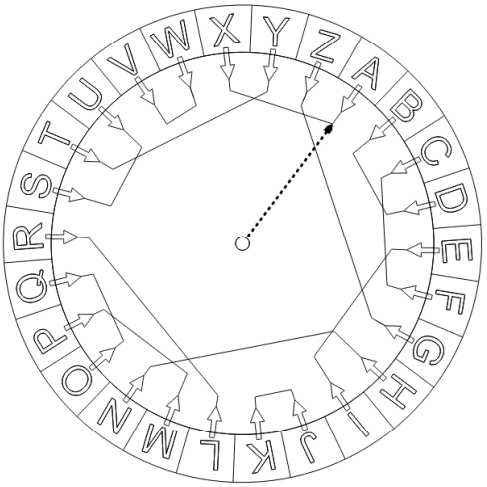
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Activity: Disc Enigma

People have used secret messages for centuries to protect important information. Even today, much of our communication online needs to be encrypted to keep our private data safe. For example, when your parents withdraw money from an ATM or make a UPI payment, they enter a PIN. This number needs to be hidden so that no one else can take their money, or when your parents buy something online and enter their bank card details, the information needs to be protected so that others cannot see or steal it.

A secret is information shared only with someone you completely trust through a message. Encryption is the process of changing a message so that its original meaning is hidden from anyone except the intended receiver. Though we use sophisticated methods to encrypt information today, we will understand encryption through a simple activity.



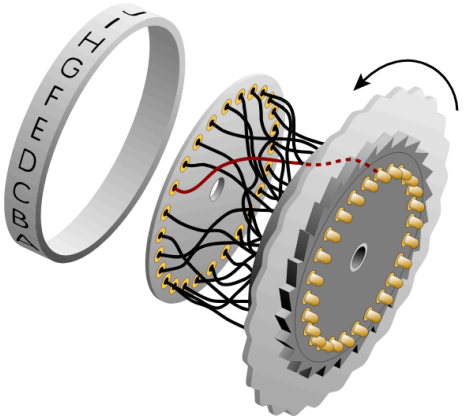


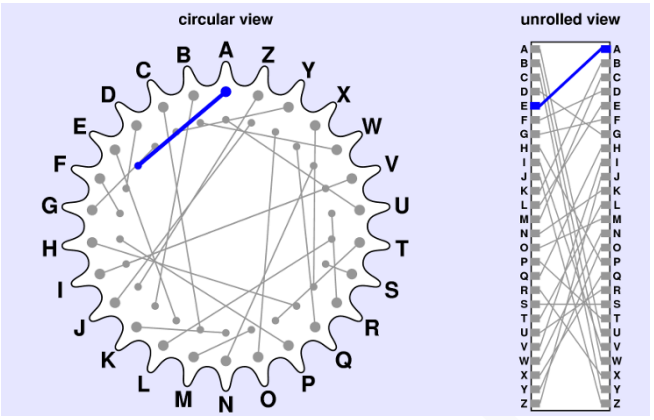
One simple way to do this is through ciphers. Cipher scrambles up the message using a **key**. To unscramble and access the message, you must have the key. This way, the cipher protects your message from those who do not have the key.

A well-known example of a cipher is the Enigma machine. Used during World War 2, it allowed for billions and billions of ways to encode a message, making it incredibly difficult to crack the code. For a time, the code seemed unbreakable. Breaking the code of which shortened the war and saved an estimated 15 million lives.

The **Disc Enigma** is a simple representation of the Enigma Machine. We can use this method to securely send important messages. In the Disc Enigma, the letters of the alphabet are shifted by a certain pattern to be replaced by a new alphabet.

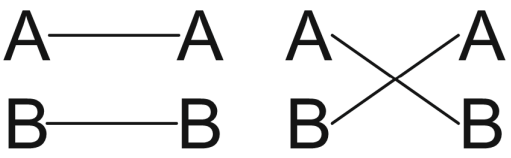
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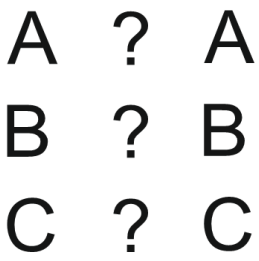


For example, if A becomes E by connecting it with the +4 pattern, and B becomes K by connecting it with the +9 pattern, every letter connects with some other letters. This simple connecting pattern method allows messages to be hidden from anyone who does not know the key, the key pattern.

If we want to connect or map the alphabet A and B to another set of alphabet A and B, there are two ways to do so. Either we can connect A to another A and B to another B, or connect A to B and B to A. This is shown here,



Q: How many ways will there be to map a set of alphabet A, B and C to another set of alphabet A, B and C?



a) 2
b) 4
c) 6
d) 8

Explanation

c) 6

We want to connect each letter in the first set (A, B, C) with exactly one letter in the second set (A, B, C). This means we are arranging the letters in different possible orders.

Below are all possible mappings:

1. A → A, B → B, C → C

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2. A → A, B → C, C → B
3. A → B, B → A, C → C
4. A → B, B → C, C → A
5. A → C, B → A, C → B
6. A → C, B → B, C → A

So there are 6 possible mappings.

Another way to see this:
Choose a mapping for A → 3 choices
Choose a mapping for B → 2 remaining choices
Choose a mapping for C → 1 remaining choice
So the total number of mappings is:
3×2×1=6
CT Competency: Pattern Recognition, Generalisation

One can extend this idea to the entire alphabet.

Making and Usage of Disc Enigma
Print out the template file on A4 Paper for each student. Make pairs of students.

Show the video demo of the Enigma Machine to the students.

Direct the Student's attention to the Video Demo of the Enigma Machine, noting that pressing a key/button to encrypt a message causes the rotor to rotate one position. After each button press, the rotors rotate, and pressing the same button routes the letter to a different connection, revealing a different letter. After each button press, a letter was mapped to another.

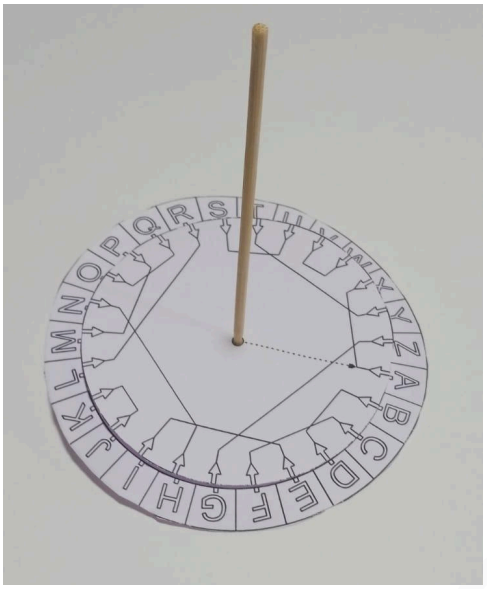
This greatly increases the number of possible configurations. Each time a key is pressed on an Enigma machine, the rotors turn, and the code changes. This means that the message "AA" could be encoded as something like "TU" even though the same key was pressed twice.

(Guide Students to Make Disc Enigma)

Instructions for Making

- Cut out both the two circles.
- Put a toothpick or a thin stick through the centre of the two circles, with the smaller one on top of the larger one.
- Make sure the smaller circle/wheel is free to rotate over the larger one.

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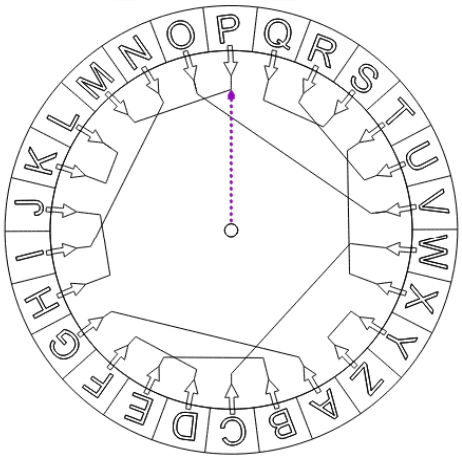


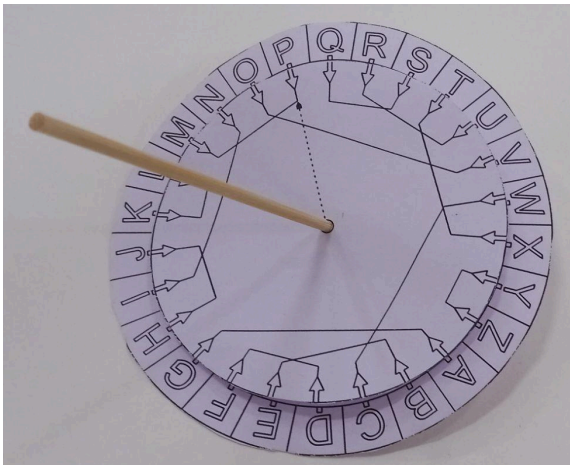
Once the students make the Disc Enigma, tell them to write a message, encrypt it, and give it to their friends.

Instructions for Usage

To Encrypt

- Choose a short message, like name, school name or the food item in the lunchbox that you would like to give to your friend, and decide on an alphabet that will represent the key alphabet.
- For example, you want to send the message “APPLE”, and you and your friend decide that ‘P’ will be the key alphabet.
- To use the Disc Enigma, first align the arrow on the smaller wheel to the key alphabet ‘P’ on the bigger circle.

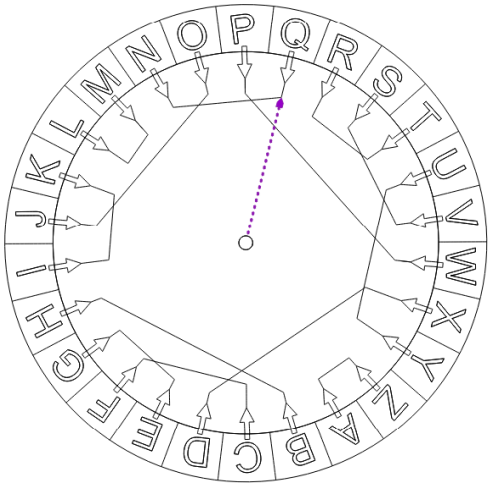


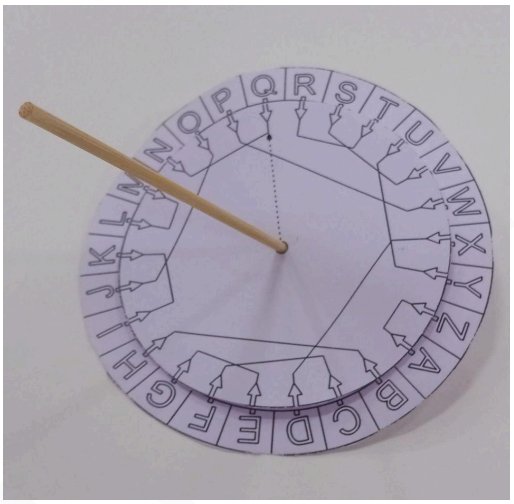


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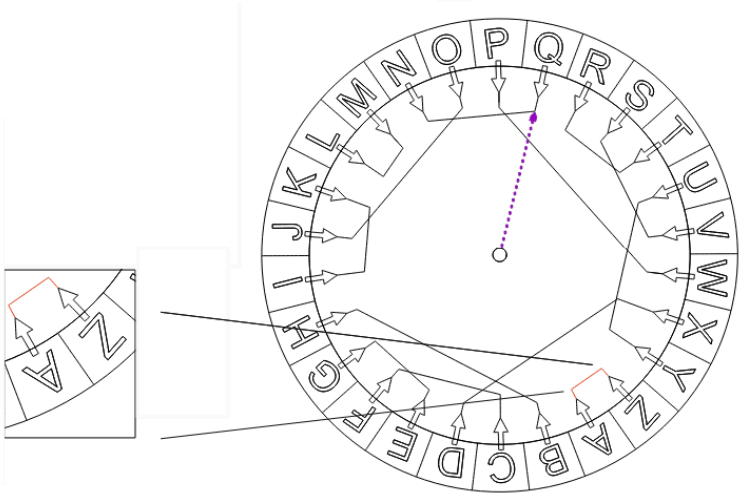
- When you encrypt a letter, you need to physically rotate the disc by one position, mimicking the rotation after each button press, as in the actual Enigma machine.

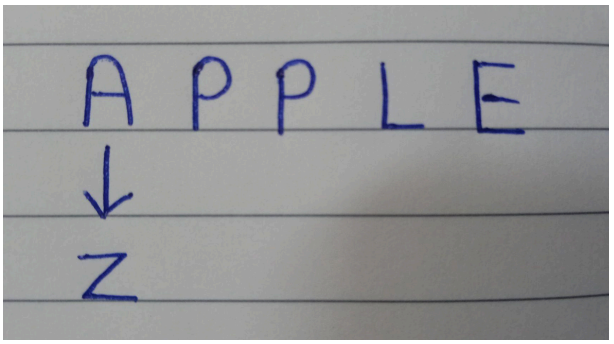
- To encode (encrypt) the first letter ‘A’ of the message “APPLE”, first rotate the arrow one position forward, from ‘P’ to ‘ Q’.





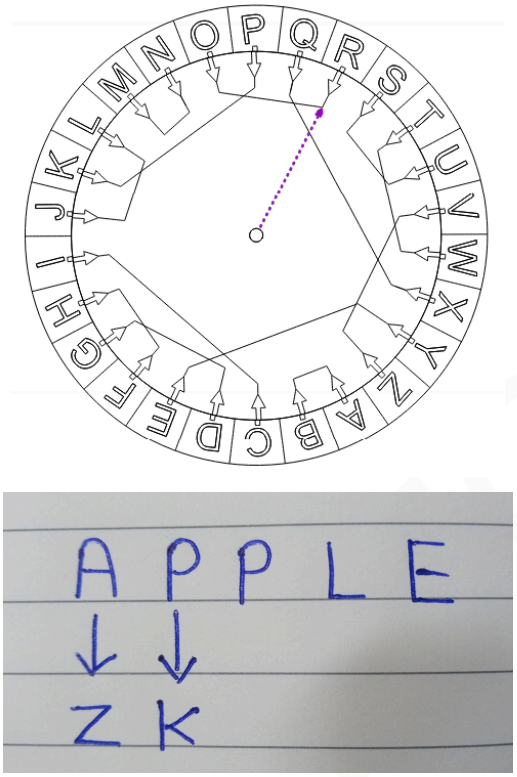
- And write the alphabet that is connected or maps to the ‘A’. Here, ‘A’ is connected to Z. Z will be written in place of A in APPLE.



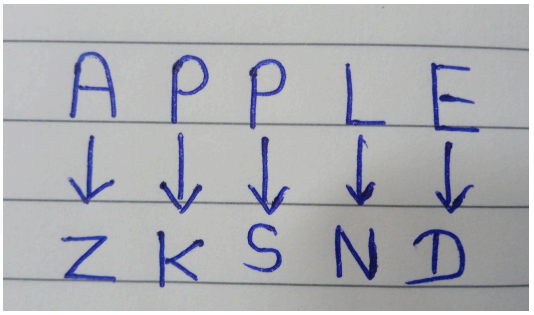


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- Now, to write the second letter ‘P’ of the message “APPLE”, again rotate the arrow one position forward, from ‘Q’ to ‘R’, and now write the alphabet that is connected or mapped to the ‘P’. Here, it will be K.



- Do this one letter at a time until the entire message is encoded. “APPLE” will be encoded as “ZKSND” with ‘P’ as the key alphabet.



- Once you have finished encoding your message, give it to your friend along with the key alphabet (key).

To Decrypt

- To decode (decrypt) the message, your friend should first align the arrow on the smaller wheel with the key letter ‘P’ on the larger circle.
- Your friend needs to follow the same steps of the encryption to decrypt the message.

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- To decode (decrypt) the first letter ‘Z’ of the message “ZKSND”, first rotate the arrow one position forward, from ‘P’ to ‘ Q’, and now write the alphabet that is connected or maps to the ‘Z’. Here, it will be A.
- A will be written in place of Z in ZKSND.
- Do this one letter at a time until the entire message is decoded. “ZKSND” will be encoded as “APPLE” with ‘P’ as the key alphabet.

Q. If the message “**HELLO**” is encoded using the key alphabet as ‘**B**’, which of the following is the correct encoded message?

a) NCMHN
b) DBJJL
c) EGREK
d) IMROZ

Explanation

c) EGREK

To encode the message using the Disc Enigma:

- First, align the arrow to the key alphabet B.
- Before encoding each letter, rotate the disc by one step.
- Then find the letter connected to the original letter on the disc.

So the process becomes:

Step	Disc Rotation	Letter to Encode	Encoded Letter
1	B → C	H	E
2	C → D	E	G
3	D → E	L	R
4	E → F	L	E
5	F → G	O	K

So the encoded message becomes: HELLO → EGREK

This happens because the disc rotates after each letter, changing the mapping each time.

CT Competency: Algorithmic Thinking

Q: How many ways will there be to connect all 26 letters to another set of 26 letters?

a) 26
b) 52!
c) 26!
d) 50

Explanation

c) 26!

Each letter of the alphabet must connect to exactly one other letter.

- The first letter has 26 choices

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- The second letter has 25 remaining choices
- The third letter has 24 choices

This continues until the last letter, which has 1 choice.
So the total number of ways is: 26×25×24×...×1 = 26!
This number is extremely large (about 4×10^{26} possibilities).
This large number of configurations is what makes encryption systems difficult to break.

CT Competency: Evaluation, Pattern Recognition, Generalisation

Think about how secure the Disc Enigma really is by exploring that there are many possible configurations. Each time a key is pressed on an Enigma machine, the rotors turn, and the code changes. This means that the message “AA” could be encoded as something like “TU” even though the same key was pressed twice. This connects directly to how modern cryptography relies on large numbers of possibilities for security.

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